90.$^+$ (for topologists) In a locally compact, connected, and locally connected Hausdorff space X, consider sequences $U_1 \supseteq U_2 \supseteq \dots$ of open, non-empty, connected subsets with compact frontiers such that $\bigcap_{i \in \mathbb{N}} \overline{U_i} = \emptyset$. Call such a sequence *equivalent* to another such sequence if every set of one sequence contains some set of the other sequence and vice versa. Note that this is indeed an equivalence relation, and call its classes the *Freudenthal ends* of X. Now add these to the space X, and define a natural topology on the extended space $\hat{X}$ that makes it homeomorphic to $|X|$ if X is a graph, by a homeomorphism that is the identity on X.

91.$^+$ Let G be a connected countable graph that is not locally finite. Show that $|G|$ is not compact, but that $\Omega(G)$ is compact if and only if for every finite set $S \subseteq V(G)$ only finitely many components of $G - S$ contain a ray.

92.$^+$ Let G be a connected graph. Assuming that G has a normal spanning tree, define a metric on $|G|$ that induces its usual topology. Conversely, use Exercise 43 to show that if $V \cup \Omega \subseteq |G|$ is metrizable then G has a normal spanning tree.

93. Find a graph G for which $|G|$ is not metrizable.

(Hint. Rather than thinking of metrics directly, recall some properties of metric spaces, and construct a graph G without such a property.)

A topological space X is *locally connected* if for every $x \in X$ and every neighbourhood U of x there is an open connected neighbourhood $U' \subseteq U$ of x. A *continuum* is a compact, connected Hausdorff space. By a theorem of general topology, every locally connected metric continuum is arc-connected.

94.$^+$ Show that, for G connected and locally finite, every connected standard subspace of $|G|$ is locally connected. Using the theorem cited above, deduce Lemma 8.6.5.

95.$^+$ Prove Lemma 8.6.6 directly, without relying on Lemma 8.6.5.

96. Let G be a locally finite graph, and X a standard subspace of $|G|$ spanned by a set of at least two edges. Show that X is a circle if and only if, for every two distinct edges $e, e' \in E(X)$, the subspace $X \setminus \mathring{e}$ is connected but $X \setminus (\mathring{e} \cup \mathring{e}')$ is disconnected.

97. Does every infinite locally finite 2-connected graph contain an infinite circuit? Does it contain an infinite bond?

98. Consider a locally finite graph.

(i) Show that every infinite circuit meets some infinite bond in exactly one edge.

(ii) Show that every infinite bond meets some infinite circuit in exactly one edge.

99. Show that the union of all the edges contained in an arc or circle C in $|G|$ is dense in C.

100.$^+$ Prove that the circle shown in Figure 8.6.1 is really a circle, by exhibiting a homeomorphism with S^1.

101. Every arc induces on its points a linear ordering inherited from $[0,1]$. Call an arc in $|G|$ *wild* if it induces on some subset of its vertices the ordering of the rationals: between every two there lies another. Show that every arc containing uncountably many ends is wild.

102. Find a locally finite graph G with a connected standard subspace of $|G|$ that is the closure of a union of disjoint circles.

103. Show that, for G locally finite, a closed standard subspace C of $|G|$ is a circle in $|G|$ if and only if C is connected, every vertex in C is incident with exactly two edges in C, and every end in C has topological degree 2.

104. Let T be a locally finite tree. Construct a continuous map $\sigma\colon [0,1] \to |T|$ that maps 0 and 1 to the root and traverses every edge exactly twice, once in each direction. (Formally: define σ so that every inner point of an edge is the image of exactly two points in $[0,1]$.)

(Hint. Define σ as a limit of similar maps σ_n for finite subtrees T_n.)

105. Let G be a connected locally finite graph. Show that the following assertions are equivalent for a spanning subgraph T of G:

(i) $\overline{T}$ is a topological spanning tree of $|G|$;

(ii) T is edge-maximal with $\overline{T}$ containing no circle;

(iii) T is edge-minimal with $\overline{T}$ a connected subspace of $|G|$.

106.$^-$ (i) Observe that a topological spanning tree need not be homeomorphic to a tree. Is it homeomorphic to the space $|T|$ for a suitable tree T?

(ii) Find a graph G with an ordinary spanning tree T whose closure in $|G|$ is not arc-connected.

107. Let T be an end-faithful spanning tree of a locally finite graph G. (Such trees are defined in Exercise 84.) Is $\overline{T}$ a topological spanning tree of $|G|$?

108. Let G be locally finite and connected, with vertices $v_0, v_1, \dots$ say. Let G_n be the minor of G obtained by contracting every component of $G - \{v_0, \dots, v_n\}$ to a vertex. Construct spanning trees T_n of G_n so that $\bigcup_n E(T_n)$ is the edge set of a topological spanning tree of G.

109. Let F be a set of edges in a locally finite connected graph $G = (V, E)$.

(i) Show that F is a circuit if and only if F is not contained in the edge set of any topological spanning tree of G and is minimal with this property.

(ii) Show that F is a finite bond if and only if F meets the edge set of every topological spanning tree of G and is minimal with this property.

110. Extend Exercise 51 of Chapter 1 to characterizations of the bonds, and of the finite bonds, in a locally finite connected graph.

111.$^+$ Prove a topological tree-packing theorem for standard subspaces X of locally finite graphs. Here, a *topological spanning tree of* X is a connected closed subspace of X that contains all its vertices but no circle.

112. To show that Theorem 3.2.6 does not generalize to infinite graphs with its finitary cycle space, construct a 3-connected locally finite planar graph with a separating cycle that is not a finite sum of non-separating induced cycles. Can you find an example where even infinite thin sums of finite non-separating induced cycles do not generate all separating cycles?

113.$^-$ As a converse to Theorem 8.7.1 (iii), show that the fundamental circuits of an ordinary spanning tree T of a locally finite graph G do not generate $\mathcal{C}(G)$ unless $\overline{T}$ is a topological spanning tree.

114.$^-$ Explain why Theorem 8.7.3 is needed in the proof of Corollary 8.7.4: can't we just combine the constituent sums of circuits for the D_i (from our assumption that $D_i \in \mathcal{C}(G)$) into one big family?

115. Deduce Corollary 8.7.4 from Theorem 8.7.1, not using Theorem 8.7.3.

116. If a finite set D of edges meets every finite cut of G evenly, must it also meet every infinite cut evenly?

117. Prove Theorem 8.7.3 by the method used to prove Theorem 8.6.9.

118. Prove Lemma 8.6.5 by the method used to prove Theorem 8.7.3.

For the next ten exercises, let G be a locally finite connected graph. Let $\mathcal{C} = \mathcal{C}(G)$, and define the *cut space* $\mathcal{B} = \mathcal{B}(G)$ of G as in Chapter 1.9. Note that cuts may now be infinite. Define 'generate' for cuts as for circuits, allowing infinite thin sums. Given a set $\mathcal{F} \subseteq \mathcal{E}(G)$, write $\mathcal{F}_{\text{fin}} := \{ F \in \mathcal{F} : |F| < \infty \}$, $\mathcal{F}^\perp := \{ D \in \mathcal{E}(G) : |D \cap F| \in 2\mathbb{N} \;\; \forall F \in \mathcal{F} \}$ and $(\mathcal{F}_{\text{fin}})^\perp =: \mathcal{F}_{\text{fin}}^\perp$.

119. Show that $\mathcal{C}$ and $\mathcal{B}$ are closed in the edge space $\mathcal{E} = \{0,1\}^E$ of G if $\{0,1\}$ carries the discrete topology and $\{0,1\}^E$ the product topology.

120. Show that the definition of $\mathcal{C}_{\text{fin}}$ given above coincides with that given in the text: that the finite elements of $\mathcal{C}$ are finite sums of finite circuits.

121. (i) Show that the fundamental circuits of any ordinary spanning tree of G generate $\mathcal{C}_{\text{fin}}$ by finite sums, but that they need not generate $\mathcal{C}$.

(ii) Show that the fundamental cuts of any topological spanning tree of G generate $\mathcal{B}_{\text{fin}}$ by finite sums, but that they need not generate $\mathcal{B}$.

122. (i)$^-$ Show that $\mathcal{B}$ is a subspace of $\mathcal{E}(G)$ generated by finite cuts.

(ii) Show that every cut is a disjoint union of bonds.

(iii)$^+$ Show that the fundamental cuts of any ordinary spanning tree of G generate $\mathcal{B}$.

(iv)$^+$ Show that $\mathcal{B}$ is closed under infinite thin sums.

123. (i)$^-$ Find in this book a proof, or sketch of a proof, for each of the following two statements: $\mathcal{C} = \mathcal{B}_{\text{fin}}^\perp$ and $\mathcal{B} = \mathcal{C}_{\text{fin}}^\perp$.

(ii)$^+$ Show that $\mathcal{B}^\perp = \mathcal{C}_{\text{fin}}$ and, if G is 2-edge-connected, $\mathcal{C}^\perp = \mathcal{B}_{\text{fin}}$.

124.$^+$ Write $\hat{\mathcal{C}}$ for the set of circuits in G, and $\hat{\mathcal{B}}$ for the set of bonds.

(i) Show that $\hat{\mathcal{C}}_{\mathrm{fin}}^{\perp} = \mathcal{C}_{\mathrm{fin}}^{\perp}$ and $\hat{\mathcal{B}}_{\mathrm{fin}}^{\perp} = \mathcal{B}_{\mathrm{fin}}^{\perp}$.

(ii) Show that every element of $\hat{\mathcal{C}}^{\perp}$ is a disjoint union of finite bonds, and that every element of $\hat{\mathcal{B}}^{\perp}$ is a disjoint union of finite circuits.

(iii) Construct 2-connected graphs with $\mathcal{C}^{\perp} \subsetneq \hat{\mathcal{C}}^{\perp}$ or $\mathcal{B}^{\perp} \subsetneq \hat{\mathcal{B}}^{\perp}$.

125. Extending Gallai's partition theorem of Exercise 55 (ii), Chapter 1, show that $E(G)$ can be partitioned into a set $C \in \mathcal{C}$ and a set $D \in \mathcal{B}$. (This strengthens Exercise 25.)

126. Let $F \subseteq E(G)$ be a set of edges.

(i)$^-$ Show that F extends to a cut if it contains no odd circuit.

(ii)$^+$ Show that F extends to some $D \in \mathcal{C}$ if it contains no odd bond.

127.$^+$ Let H be an abelian group. The group $\mathcal{C}_H$ of *H-circulations* on $|G|$ consists of the maps $\psi\colon \vec{E} \to H$ that satisfy (F1) and $\psi(X,Y) = 0$ for any finite cut $E(X,Y)$ of G. (See Chapter 6.1 for notation.) Extend Exercise 8 of Chapter 6 to $\mathcal{C}_H$, with $\mathcal{E}_H$ and $\mathcal{D}_H$ as defined there.

128. Let X be a connected standard subspace of $|G|$. Call a continuous map $\sigma\colon S^1 \to X$ a *topological Euler tour* of X if it traverses every edge in $E(X)$ exactly once. (Formally: every inner point of an edge in $E(X)$ must be the image of exactly one point in S^1.) Show that X admits a topological Euler tour if and only if $E(X) \in \mathcal{C}(G)$.

129.$^+$ An *open Euler tour* in an infinite connected graph G is a 2-way infinite walk $\dots e_{-1} v_0 e_0 \dots$ that contains every edge of G exactly once. Show that G contains an open Euler tour if and only if G is countable, every vertex has even or infinite degree, and any finite cut $F = E(V_1, V_2)$ with both V_1 and V_2 infinite is odd.

130. By Exercise 23 of Chapter 4, every finite 2-connected graph without a K^4 or $K_{2,3}$ minor contains a *Hamilton* cycle, one that contains all its vertices. Show that every locally finite such graph has a *Hamilton circle*, a circle in $|G|$ containing all the vertices (and ends) of G.

131.$^+$ Extend Theorem 2.4.4 to packings and coverings of locally finite graphs with topological spanning trees in appropriate spaces obtained from $|G|$.

132. Where in the proof of Theorem 8.8.2 do we use that G is connected?

133. Use the techniques from Section 8.8 to prove that the wild circle is indeed a circle. Your proof may be informal in its handling of the wild circle graph G – use a picture rather than its formal definition.

134.$^+$ Use the methods from Section 8.8 to prove that connected standard subspaces of $|G|$, for G locally finite, have topological spanning trees: closed connected subspaces containing all their vertices.

135.$^+$ Consider the space $\|G\|$ for the edgeless countably infinite graph G. Have you met this space before in another guise?

Notes

There is no comprehensive monograph on infinite graph theory, but over time several surveys have been published. A relatively wide-ranging collection of survey articles can be found in R. Diestel (ed.), *Directions in Infinite Graph Theory and Combinatorics*, North-Holland 1992. (This has been reprinted as Volume 95 of the journal *Discrete Mathematics*.) Some of the articles there address purely graph-theoretic aspects of infinite graphs, while others point to connections with other fields in mathematics such as differential geometry, topological groups, or logic. A similar collection, edited by R. Diestel, B. Mohar and G. Hahn, appeared in 2011 as a special issue of *Discrete Mathematics (vol. 311)*. It includes a survey of everything to do with $|G|$, the topological space consisting of an infinite graph G and its ends: R. Diestel, Locally finite graphs with ends: a topological approach, arXiv:0912.4213. Another very instructive survey in this volume is by L. Soukup, Elementary submodels in infinite combinatorics, arXiv:1007.4309.

A survey of infinite graph theory as a whole was given by C. Thomassen, Infinite graphs, in (L.W. Beineke & R.J. Wilson, eds.) *Selected Topics in Graph Theory 2*, Academic Press 1983. This also treats a number of aspects of infinite graph theory not considered in our chapter here, including problems of Erdős concerning infinite chromatic number, infinite Ramsey theory (also known as *partition calculus*), and reconstruction. The first two of these topics receive much attention also in A. Hajnal's chapter of the *Handbook of Combinatorics* (R.L. Graham, M. Grötschel & L. Lovász, eds.), North-Holland 1995, which has a strong set-theoretical flavour. Péter Komjáth is currently preparing a monograph about this kind of infinite graph theory. A specific survey on reconstruction by Nash-Williams can be found in the *Directions* volume cited above. R. Halin, Miscellaneous problems on infinite graphs, *J. Graph Theory* **35** (2000), 128–151, contains his legacy of unsolved infinite graph problems.

Halin's book, *Graphentheorie* (2nd ed.), Wissenschaftliche Buchgesellschaft 1989, is also good general reference for infinite graphs. A more specific monograph about *simplicial decompositions* of infinite graphs (see Chapter 12) is R. Diestel, *Graph Decompositions*, Oxford University Press 1990. Our Chapter 12.6 closes with two theorems about forbidden minors in infinite graphs.

When sets get bigger than countable, combinatorial set theory offers some interesting ways other than cardinality to distinguish 'small' from 'large' sets. Among these are the use of *clubs* and *stationary sets*, of *ultrafilters*, and of *measure and category*. See P. Erdős, A. Hajnal, A. Máté & R. Rado, *Combinatorial Set Theory: partition relations for cardinals*, North-Holland 1984; W.W. Comfort & S. Negropontis, *The Theory of Ultrafilters*, Springer 1974; J.C. Oxtoby, *Measure and Category: a survey of the analogies between topological and measure spaces* (2nd ed.), Springer 1980.

Infinite matroids, whose study long lay dormant for want of a clear idea of what exactly they should be, were finally axiomatized in H. Bruhn, R. Diestel, M. Kriesell & P. Wollan, Axioms for infinite matroids, *Adv. Math.* **239** (2013), 18–46, arXiv:1003.3919. Since that paper, the theory of infinite matroids has flourished. It draws much inspiration from topological infinite graph theory as presented in Sections 8.6–8.7, but also has its own problems and agenda. Regular updates can be found in Nathan Bowler's blog at